MIL-E-1/1472C
20 February 1976
SUPERSEDING
MIL-E-1/1472B
24 November 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8188

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode
See figure 1
Mounting position: See note 6
Weight: 9 ounces (255.15 grams) nominal

ABSOLUTE RATINGS: F = 110 MHz

Parameter: Unit:	Ef Vac (see note 1)	eb kv	Eb kVdc	Ec1 Vdc	Ec2 Vdc	Ib mAdc	ib a	ik a
Maximum:								
Pulse mod: (R load)	5.0±5%	---	20	-1,000	2,500	---	4.0	6.0
Anode pulsed osc or amp:	5.0±5%	15	---	-1,000	2,500	---	---	6.0
Grid pulsed osc or amp:	5.0±5%	---	10	-1,000	2,500	---	---	6.0
TEST CONDITIONS:	5.0	---	2.5	Adj	500	160	---	---

ABSOLUTE RATINGS: F = 110 MHz

Parameter: Unit:	Pp W	Pg2 W	Pg1 W	TE °C	T(seal) °C (see note 2)	Du --- (see note 3)	Cooling --- (see note 4)
Maximum:							
Pulse mod: (R load)	400	35	10	225	200	---	---
Anode pulsed osc or amp:	400	35	10	225	200	---	---
Grid pulsed osc or amp:	400	35	10	225	200	---	---
TEST CONDITIONS:	---	---	---	---	---	---	See note 5

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	13.7	14.9	Aac
1231	Peak emission	-	eb = ec1 = ec2 = 2,500 v	0.65	II	is	7.0	---	a
1266	Total grid current	-		0.65	II	Ic1	---	-10	μAdc
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-55	-80	Vdc
1266	Primary-grid emission (control)	-	Ef = 5.5 Vac; t = 15; Ic1 = 165 mAdc; anode and screen grid floating	0.65	II	Isgp	---	-100	μAdc
1266	Primary-grid emission (screen)	-	Ef = 5.5 Vac; t = 15; Ic2 = 135 mAdc; Ec1 = 0; anode floating	0.65	II	Isg2	---	-200	μAdc
1261	Electrode voltage (grid)	-	Eb = 20 kVdc; Ec2 = 1,500 Vdc; Ec1/Ib = 10 μAdc	0.65	II	Eco	---	-675	Vdc
---	High-voltage operation	7	Eb = 25 kVdc; Ec2 = 2,500 Vdc; Ec1 = -1,000 Vdc (max); tp = 10 μs; prr = 2,000; egk/ib = 0.6 a	0.65	II	---	---	---	---
	Quality conformance inspection, part 2	8							
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
2214	RF useful output power	-	Eb = 3,000 Vdc; Ib = 350 mAdc; Ec2 = 500 Vdc; F = 110 MHz (min)	---	---	Po	500	---	W (useful)
1372	Current division (method B, short pulse)	-	Eb = 2,000 Vdc; Ec2 = 1,500 Vdc; Ec1 = -500 Vdc; tp = 15 μs (max); egk/ib = 4a	---	---	egk	---	200	v
						ic1	---	150	ma
						ic2	---	500	ma
1331	Direct-interelectrode capacitance	-		---	---	Cgp	---	0.17	pF
						Cin	10.7	14.5	pF
						Cout	4.2	5.6	pF
	Quality conformance inspection, part 3								
---	Life test	9	Group C; RF useful output power; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	5.6	---	a
1266	Primary - grid emission (control)	-		---	---	Isg1	---	-100	μAdc
1266	Primary - grid emission (screen)	-		---	---	Isg2	---	-200	μAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
1042	Shock, specified pulse	8, 10	No voltages applied; accel = 15 G peak (min); D = 11 ± 2 ms half-sine wave	---	---	---	---	---	---
---	Shock, specified pulse end points:								
1261	Electrode voltage (grid)	-		---	---	Ec1	-55	-80	Vdc
1266	Total grid current	-		---	---	Ic1	---	-10	μAdc

NOTES:

1. Filament voltage shall be maintained as closely as possible, since the peak current capability of the tube is dependent on filament voltage. A decrease in peak currents may be expected if the voltage falls below normal, while higher voltages may result in decreased tube life. Filament voltage shall be measured directly at the socket.

2. It is recommended that a heat-dissipating connector (Eimac type HR-6, or equivalent) be used on the anode terminal in all rf applications.

3. During operation under pulsed conditions, duty shall be maintained at a low enough level that average tube element dissipation ratings are not exceeded. For pulse lengths in excess of 0.1 second, sone reduction of element dissipation ratings will be required. During anode pulsed operation, for applied eb values in excess of 10 kv, the maximum allowable pulse length is 10 μs.

4. Adequate forced-air cooling shall be provided to maintain base and anode seal temperatures below their maximum ratings. In all cases of operation the use of the heat-dissipating connector is recommended and a socket and chimney should be employed which provide for cooling of the seals and envelope. The following table applies to the Eimac SK-400 and SK-410 sockets, or equivalent, with the SK-406 chimney, or equivalent, at sea level and with air at 25°C, and will be considered as minimum requirements. The approximate pressure-drop values given for the SK-400, or equivalent, are as measured in the socket; those for the SK-410, or equivalent, are as measured in the equivalent of a pressurized chassis or plenum chamber. The data applies with maximum rated anode dissipation.

	SK-400 SK-406, or equivalent		SK-410 SK-406, or equivalent	
	Airflow (cfm)	Approx press drop (In. H_2O)	Airflow (cfm)	Approx. press drop (In. H_2O)
... to 110 MHz	21.0	1.20	21.0	0.60
Below 30 MHz	12.0	0.38	12.0	0.18

Cooling air shall be supplied before or simultaneously with the application of filament voltage and may be removed simultaneously with filament voltage. Operation at higher altitudes or higher ambient temperature will require an increase in cooling airflow.

5. In all electrical tests involving application of filament voltage, forced-air cooling is permitted at the rate of 21.0 cfm maximum, using the SK-400 or SK-410 and SK-406 chimney, or equivalents.

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NOTES: -Continued

6. The tube shall be mounted vertically, base down or base up. The metal tube-base shell should be grounded by means of suitable spring fingers. The tube shall be protected from severe shock and vibration.

7. During a test period of 5 minutes, the tube shall operate without arcing. Use circuit shown on figure 2, or equivalent.

8. When type 8438 of MIL-E-1/887 and type 8188 are physically identical, and have been made in the same production run, differing only in the high-voltage processing and testing of the type 8188, the one sample may represent both types insofar as the listed tests are identical.

9. When type 8438 of MIL-E-1/887 and type 8188 are physically identical, and have been made in the same production run, differing only in the high-voltage processing and testing of the type 8188, then one life-test sample may represent both types.

© 10. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

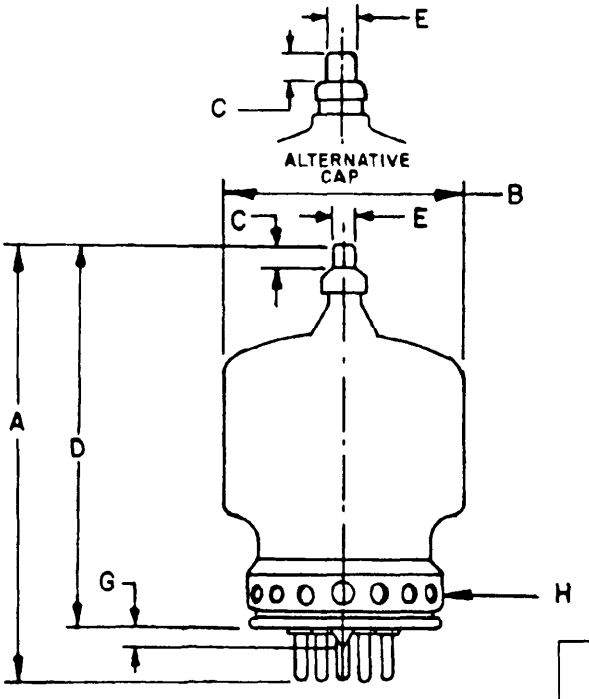
User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

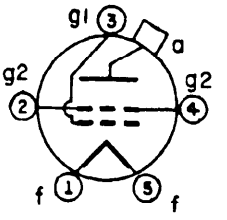
Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.875 (149.23)	6.375 (161.93)
B		3.563 (90.50)
G		.250 (6.35)
© Quality conformance inspection, part 3 (see note b)		
C	.328 (8.33)	
D	5.125 (130.18)	5.625 (142.88)
E	.350 (8.91)	.365 (9.27)
H	Base: A5-97 EIA	

NOTES:

a. Base pins T and tubulation K are so alined that they can be freely inserted in a gage .250 (6.35 mm) inch thick with hole diameters of .204 (5.18 mm) and .500 (12.70 mm) inch, respectively, located on the true centers by the given dimensions S, U, V.

b. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL of 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

© FIGURE 1. Outline drawing of electron tube type 8188.

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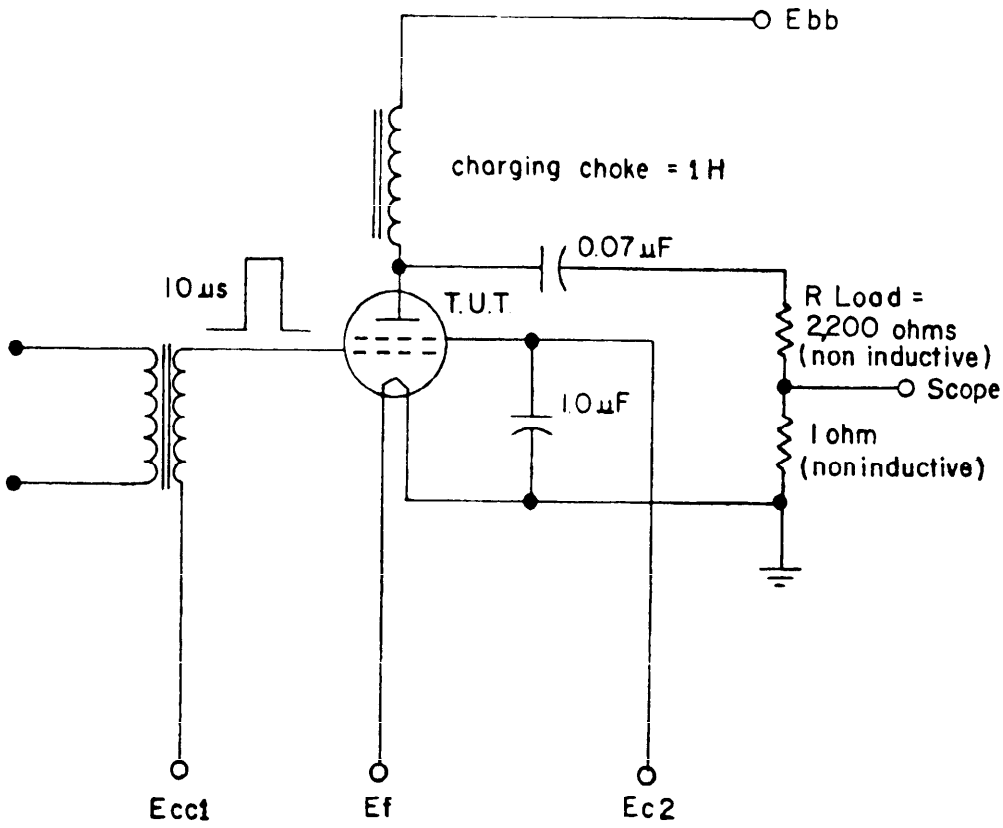


FIGURE 2. High-voltage operation test circuit.

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